FLOODGAUGE · DESIGN DECISION RECORD

FloodGauge recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	FLG-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/decisions/0002-recommendations-accepted.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 and O3 decided by Amish on 2026-10-02 (FLG-DEC-001, items 5 and 6)
0.2	2026-09-26	Amish Chadha	Budget approved by Amish ($160)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below with a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation stayed "Proposed, awaiting Amish" at this record; O1 and O3 were decided by Amish on 2026-10-02 (FLG-DEC-001, items 6 and 5): "i approve your recommendations for all 555 open decisions."

Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." At that point this repo had nine items adopted for TRL 3 work pending his review (FLG-DDR-001, D1 to D9), one open item with no recommendation (O1), and five further items raised by the TRL 3 calculations in `docs/REVIEW.md` (session 2026-09-25, TRL 3, "Still awaiting Amish", items 2 to 6). Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and nothing here goes past TRL 3.

Options considered

The options for each item are those in `docs/REVIEW.md` (both 2026-09-25 sessions), FLG-DDR-001 and FLG-CAL-001 v0.1.

Decision

Table 1. Items from FLG-DDR-001, now decided.

#	Item	Decision	What changed in the repo
D1	Budget	FieldNode costed in its own repo; FloodGauge-specific parts $150 or less	`budget_usd` stays $150 (the recommended figure); R16 wording as in FLG-REQ-001 v0.3
D2	Heads	Street and drain heads on one node	None beyond the status wording
D3	Street sensing	60 GHz radar; ultrasonic as the lower-cost variant	None beyond the status wording
D4	Drain head	Ultrasonic in a slotted stilling tube with a wet probe	None beyond the status wording
D5	Head position	Arm over the gutter, height to the clearance rule	Height now set under N2 below
D6	Alert bands	150 mm and 300 mm plus drain-full, set with the partner city	First band below 150 mm added under N5
D7	Network	Private or city gateway for events; TTN for pilots only	None beyond the status wording
D8	Privacy	Levels only	None beyond the status wording
D9	Pitch and problem	No change	`project.yaml` and `README.md` keep the existing wording

All nine: Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items raised at TRL 3, now decided.

#	Item	Decision	What changed in the repo
N1	R3 wording	Restate R3 as "continuous reading from 50 mm above the basin floor to 330 mm below the road, plus a drain-full signal above that"	FLG-REQ-001 v0.4; R3 not met to met on paper
N2	Head height and R1	Option (a): widen the R1 band to 2.5 to 5.0 m and set the height per site to the road authority's clearance rule, about 4.6 m where trucks use the curb lane	R1 target in FLG-REQ-001 v0.4; lens 2.95 m to 4.6 m and pole 3.3 m to 4.9 m in `cad/src/model.py`; FieldNode center 2.35 m to 3.0 m and street cable 2 m to 3 m to keep the I2C bus under 400 pF (350 pF); FLG-DWG-001 Rev P1 to P2
N3	R12 cable route	Option (b) for pilots: out at the grate frame and along the curb face under a bolted steel cover; option (a), the conduit, at permanent sites laid with road works	BOM line 7 conduit ($8.00) to surface cable cover ($14.00); model item 7 redrawn; installation with no civil work, 120 min; R12 still not met on time
N4	Arm anti-rotation	Add a through-bolt or pinned clamp; the through-bolt is used	M8 A4 through-bolt at the lower clamp in the model and BOM line 3 ($18.00 to $20.00); twist factor 1.45 to 37
N5	First alert band below 150 mm	The alert service supports a first band below 150 mm; its depth is set with the partner city under D6	FLG-PRC-001 v0.4 and FLG-REQ-001 v0.4 text; no depth chosen, because none was recommended

All five: Decided by Amish, 2026-09-25: go with recommendation.

Table 3. Items still open.

#	Item	Status
O1	Pilot partner and city for co-design and a first site. No preference stated and no recommendation made. The depth of the N5 band waits on this.	Decided by Amish, 2026-10-02 (FLG-DEC-001, item 6): a city or county flood agency with street flooding at grated curb inlets and an existing gauge or alert programme; first candidate type to approach, a county flood control district such as the Harris County Flood Control District; first alert band 100 mm unless the partner's practice says otherwise. Nothing is agreed.
O2	R16 over budget after N3 and N4: $153.50 against $150 (new, raised by FLG-CAL-001 v0.2)	Decided by Amish, 2026-09-26: `budget_usd` $160 (see below)
O3	R12 installation 120 min against 90 min on the surface route (new)	Decided by Amish, 2026-10-02 (FLG-DEC-001, item 5): R12 relaxed to 120 min for the pilot surface route; 90 min stays the target for permanent sites on the conduit route.

Consequences

- FLG-REQ-001 v0.4, FLG-PRC-001 v0.4, FLG-CAL-001 v0.2 and FLG-DDR-001 v0.2 carry the new wording and numbers. FLG-DWG-001 is at Rev P2. The STEP, STL and media files are regenerated from the model.
- Requirement status: 2 not met (R12, R16), 3 at risk (R7, R9, R14), 1 not verifiable at TRL 3 (R10), 8 met on paper, 3 met by design. R3 moves from not met to met on paper; R16 moves from met on paper to not met.
- Cost: FloodGauge-specific parts $145.50 to $153.50; complete gauge $271.50 to $279.50. `budget_usd` stays $150.
- Mass on the pole 4.62 kg to 4.74 kg; still under 5.0 kg.
- A site now needs a pole about 4.9 m tall above the sidewalk where tall vehicles use the curb lane, and a mobile elevating platform to fit the arm.
- Cross-repo actions are listed in `docs/REVIEW.md`; no other repo was edited.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building, testing or purchasing.

Budget approved, 2026-09-26

On 2026-09-26 Amish wrote, in chat: "i approve all the budget items."

- Budget set to $160 to cover the priced BOM: decided by Amish, 2026-09-26. This settles O2. FloodGauge-specific parts are $153.50 (lines 2 to 10; the $126.00 FieldNode core stays costed in its own repo under D1), so R16 moves from not met to met on paper with a $6.50 margin.
- Requirement status: 1 not met (R12), 3 at risk, 1 not verifiable at TRL 3, 9 met on paper, 3 met by design.
- Files changed: `project.yaml` (`budget_usd` 150 to 160); FLG-REQ-001 v0.5; FLG-CAL-001 v0.3, `docs/04-calcs/sizing.py` (target read from `budget_usd`) and `results.csv`; FLG-PRB-001 v0.5 and FLG-PRC-001 v0.5 (budget figure); `README.md`; `bom/bom-notes.md`; `docs/REVIEW.md`.